Ring Algorithm

Group 9

CSE-A

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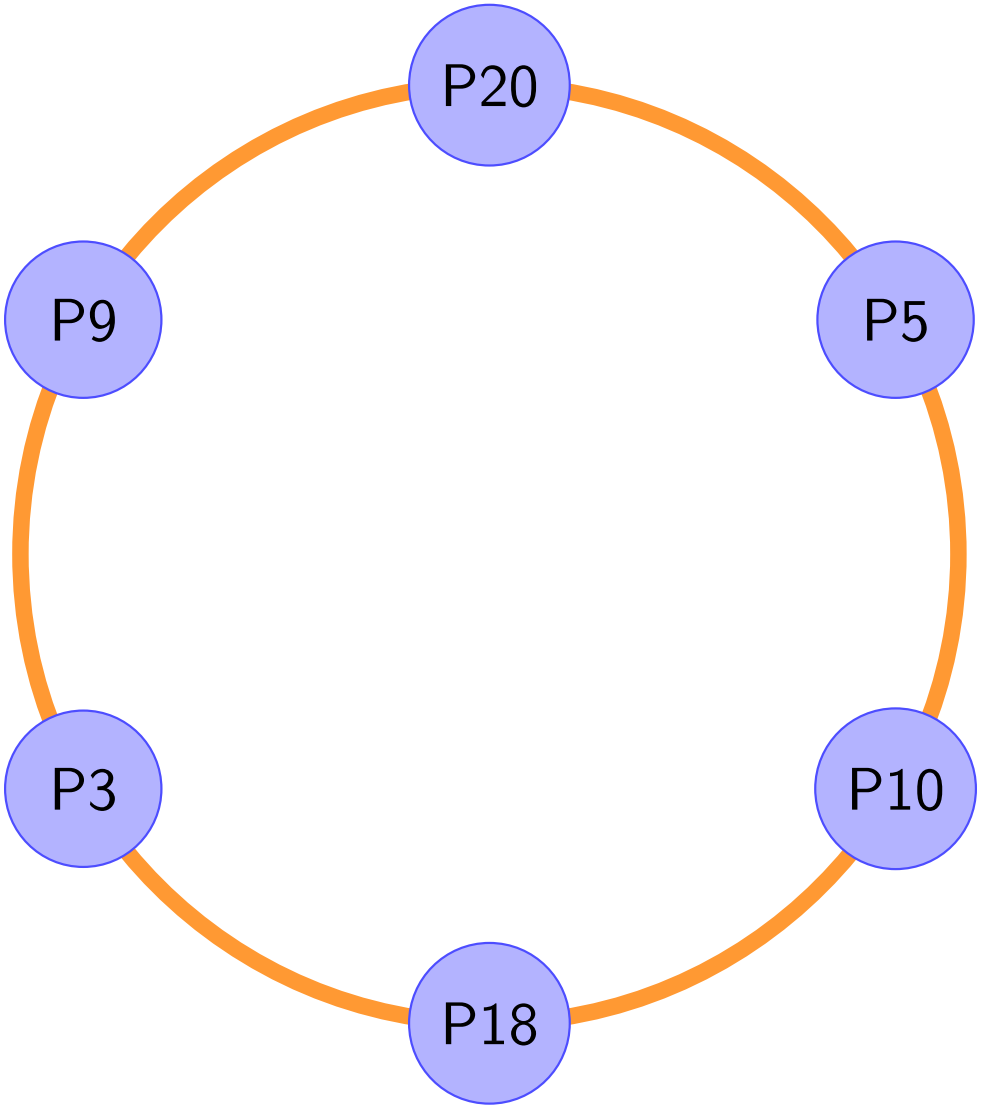
Election Algorithm Overview

- An **election algorithm** is used to choose a unique process to act as a **coordinator**.
- All processes in the system must **agree on the selected coordinator**.
- If the current coordinator retires or fails, a **new election** is initiated.
- A process that initiates an election is said to **call the election**.
- At any time, a process can be:
  - **Participant** – currently involved in an election
  - **Non-participant** – not involved in an election
- Two common election algorithms are:
  - Ring-based Election Algorithm
  - Bully Algorithm

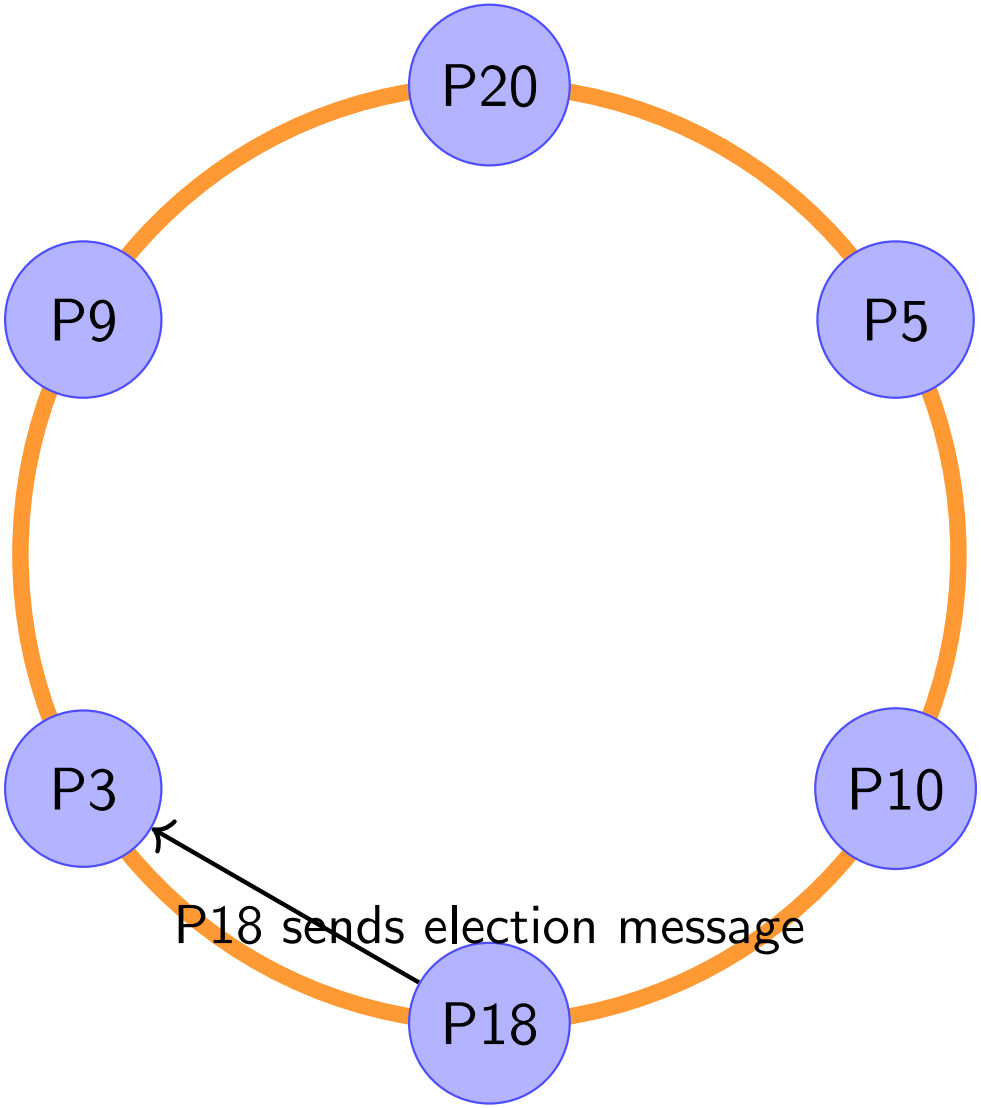
Ring-Based Election Algorithm

- Processes are arranged in a **logical ring**.
- Each process P_i communicates only with $P_{(i+1) \bmod N}$.
- All messages move **clockwise** around the ring.
- Initially, all processes are marked as **non-participants**.
- Any process can start an election by:
  - Marking itself as a participant
  - Sending an **election message** containing its identifier to its neighbour
- When a process receives an election message:
  - If received ID is **greater**, forward the message
  - If received ID is **smaller** and receiver is not a participant, replace it with its own ID and forward
  - If already a participant, it does not forward the message
- If a process receives its **own ID**, it becomes the **coordinator**.
- The coordinator sends an **elected message** around the ring to announce itself.

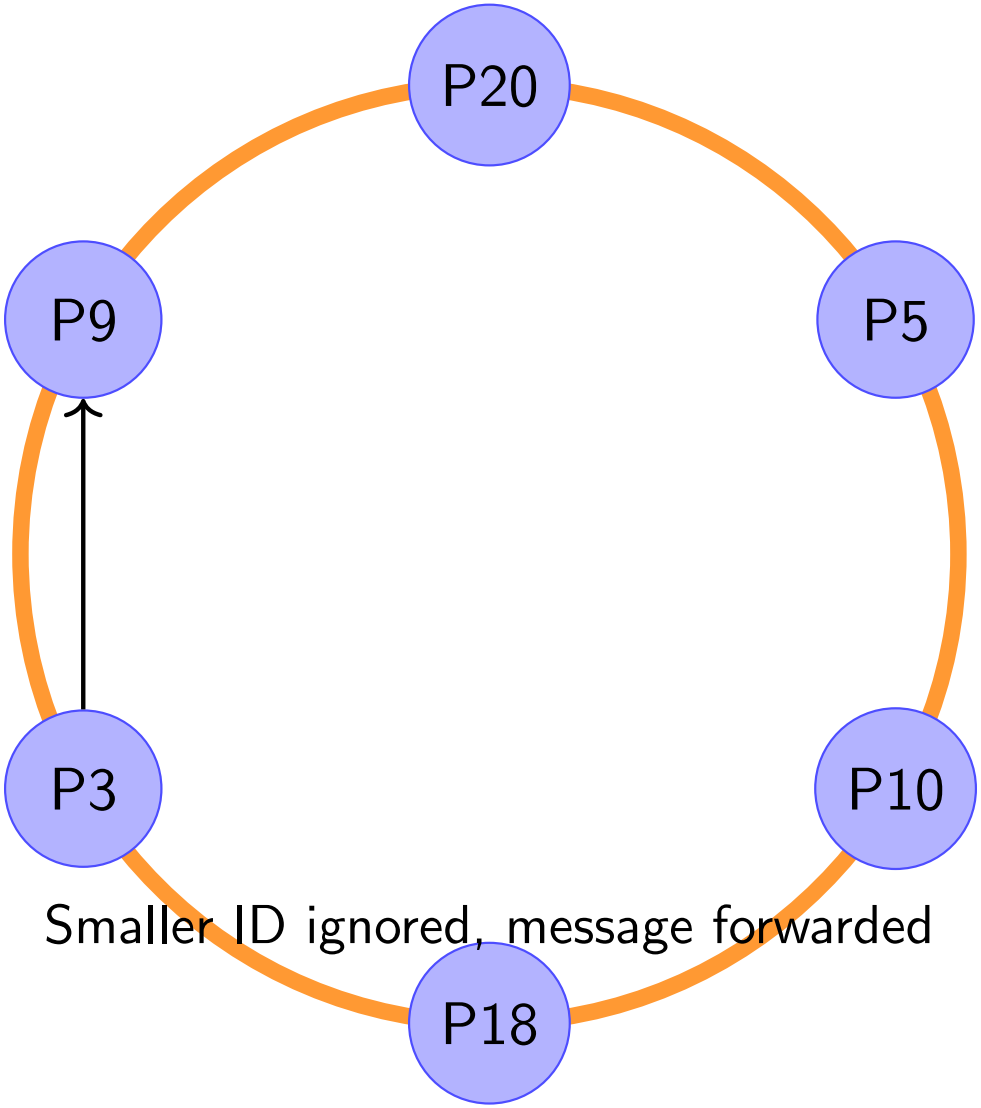
Ring Algorithm – Step 1: Initial State



Ring Algorithm – Step 2: Election Initiation

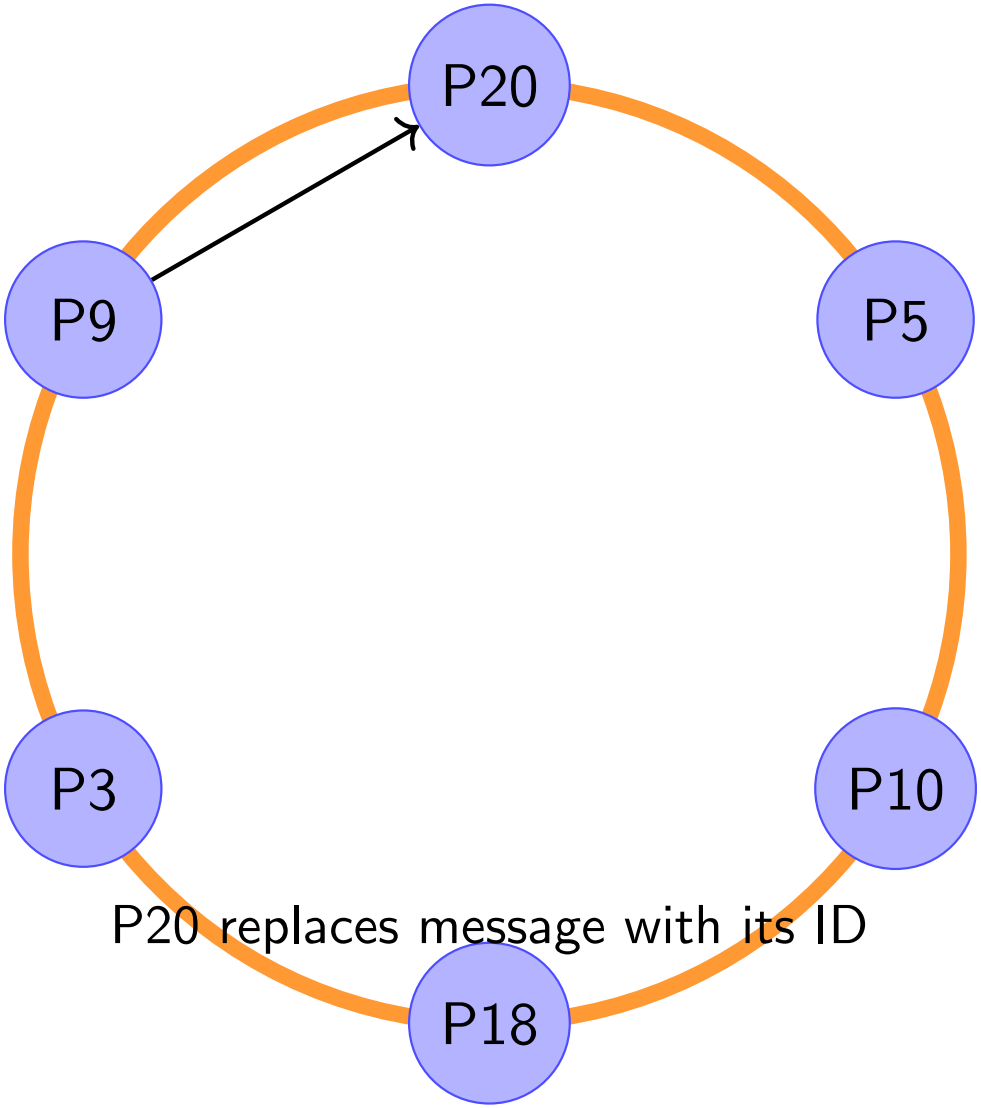


Ring Algorithm – Step 3: ID Comparison



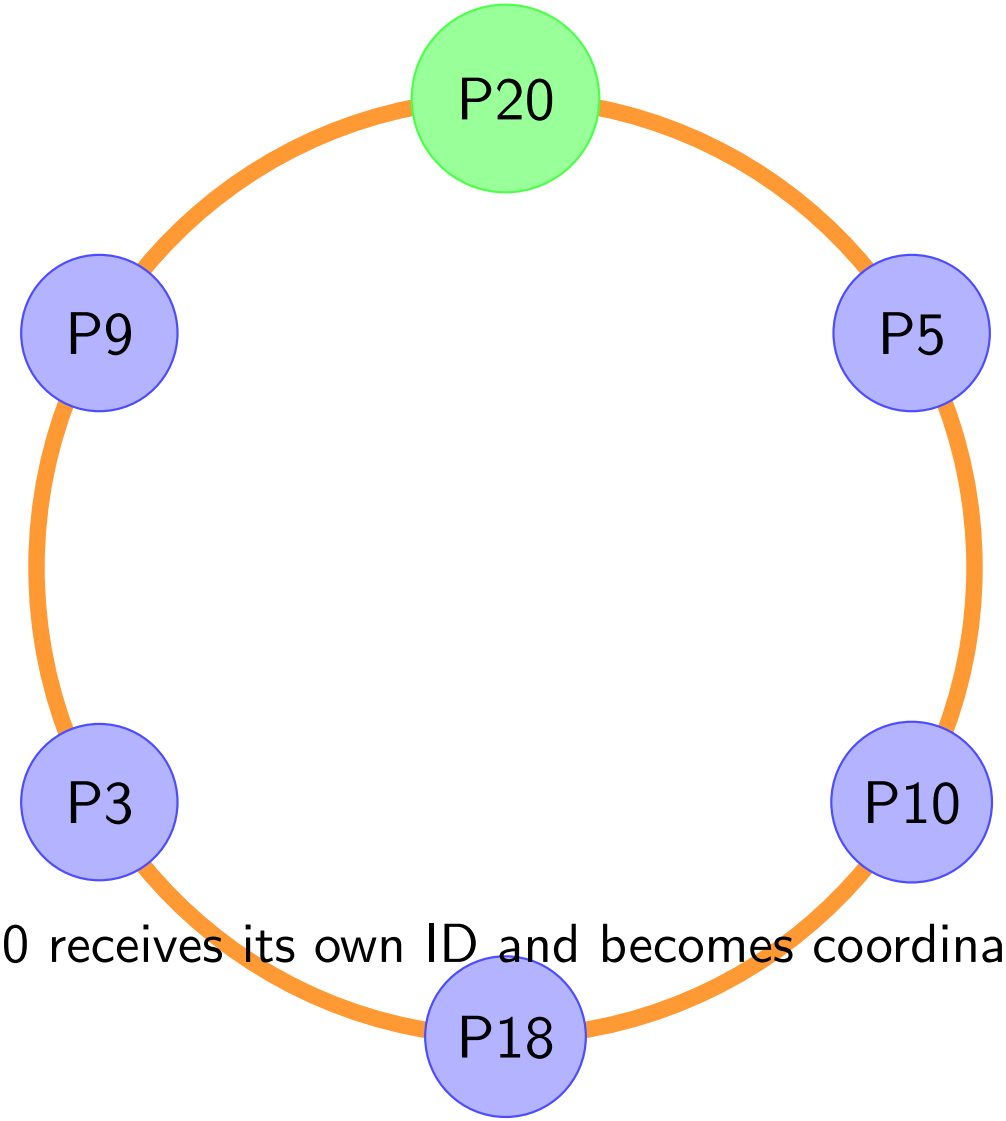
Smaller ID ignored, message forwarded

Ring Algorithm – Step 4: Higher ID Replacement



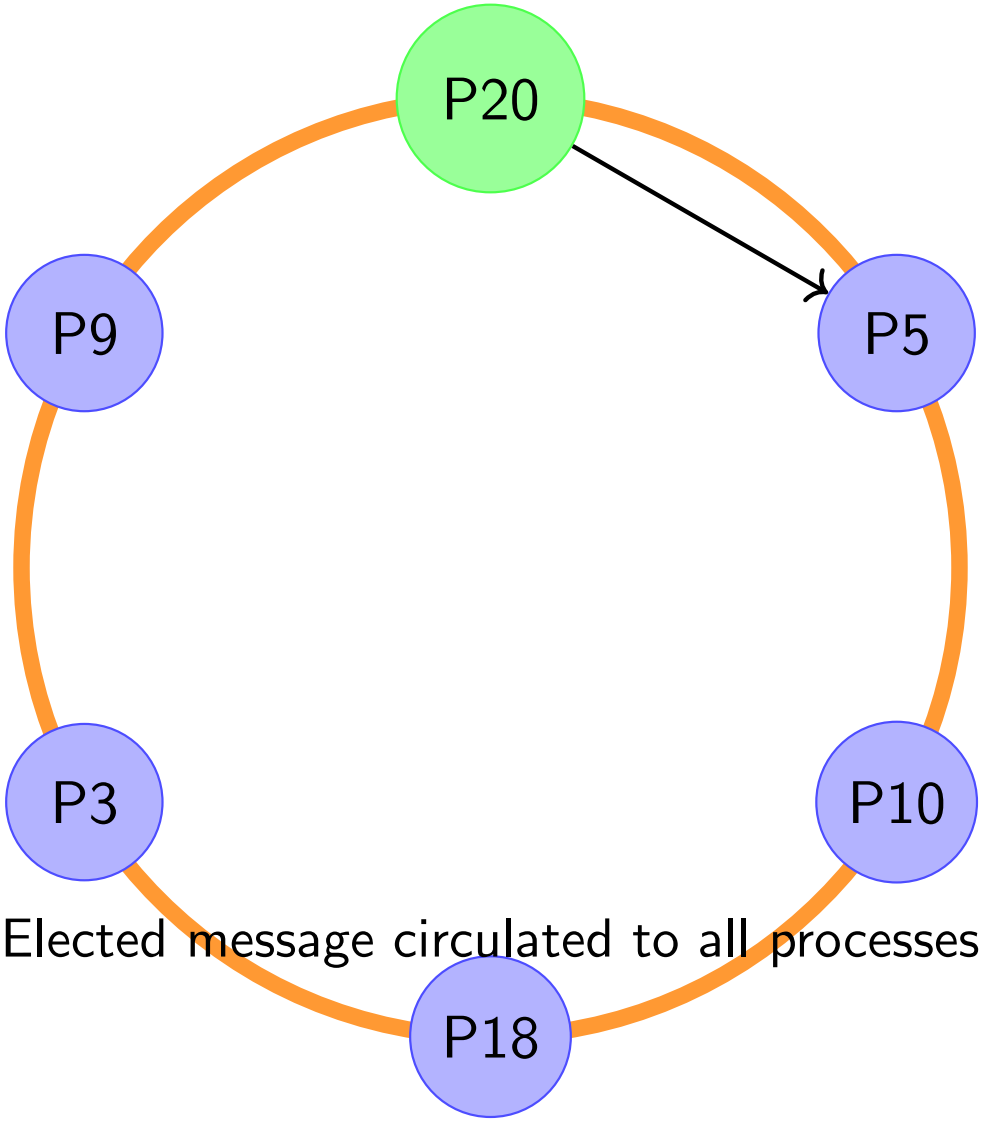
P20 replaces message with its ID

Ring Algorithm – Step 5: Coordinator Election



P20 receives its own ID and becomes coordinator

Ring Algorithm – Step 6: Coordinator Announcement



Elected message circulated to all processes

Need of Ring Algorithm in IoT Networks

- IoT networks consist of multiple **sensors and smart devices**.
- One device must act as a **coordinator / cluster head**.
- If the coordinator fails, the system must:
  - Detect failure
  - Elect a new leader automatically
- Ring Algorithm ensures:
  - Fault tolerance
  - Continuous network operation
  - Distributed control (No central dependency)

Importance in IoT Systems

- Suitable for **low-power IoT devices**.
- Requires **minimal memory and computation**.
- Simple message passing mechanism.
- Efficient for **small and medium IoT clusters**.
- Ensures agreement among all nodes on selected coordinator.

Real-Time Applications in IoT

- **Industrial IoT** – Machine sensors elect cluster head.
- **Smart Cities** – Traffic controllers elect coordinator.
- **Healthcare IoT** – Wearable devices select master node.
- **Smart Agriculture** – Soil sensors elect data aggregator.
- **Smart Homes** – Devices elect hub if main hub fails.

Requirements to Implement in IoT System

- **Hardware Requirements:**
  - Microcontroller (ESP32 / Arduino / Raspberry Pi)
  - Network module (Wi-Fi / ZigBee / LoRa)
- **Software Requirements:**
  - Embedded C / Python
  - Communication protocol (MQTT / TCP-IP)
  - Failure detection (Heartbeat mechanism)
- Each node must:
  - Know its next neighbour
  - Have a unique identifier
  - Support reliable message passing

Implementation of Ring Election Algorithm

- Implemented using **C programming**.
- User inputs:
  - Number of processes
  - Process IDs
  - Index of initiator
- Election message circulates in logical ring.
- Highest Process ID becomes **Coordinator**.
- Coordinator sends **ELECTED message** to all processes.
- Output shows:
  - Election message passing
  - Coordinator selection
  - Final announcement

C Implementation of Ring Algorithm

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i, initiator;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int pid[n];

    printf("Enter process IDs:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &pid[i]);

    printf("Enter index of initiator (0 to %d): ", n-1);
    scanf("%d", &initiator);

    int max = pid[initiator];
    int current = initiator;

    printf("\n===== ELECTION STARTED BY PROCESS %d =====\n",
            pid[initiator]);
```

C Implementation of Ring Algorithm

```
    do {
        int next = (current + 1) % n;
        printf("Process %d -> Process %d : ELECTION(%d)\n",
                pid[current], pid[next], pid[current]);
        if(pid[next] > max)
            max = pid[next];
        current = next;

    } while(current != initiator);
    printf("\n***** PROCESS %d BECOMES COORDINATOR *****\n",
            max);
    printf("\n===== COORDINATOR ANNOUNCEMENT =====\n");
    current = initiator;
    do {
        int next = (current + 1) % n;
        printf("Process %d -> Process %d : ELECTED(%d)\n",
                pid[current], pid[next], max);
        current = next;
    } while(current != initiator);
    printf("\n===== ELECTION COMPLETED =====\n");
    return 0;
}
```

Program Output

```
Enter number of processes: 4
Enter process IDs:
2
3
5
6
Enter index of initiator (0 to 3): 0

===== ELECTION STARTED BY PROCESS 2 =====

Process 2 -> Process 3 : ELECTION(2)
Process 3 -> Process 5 : ELECTION(3)
Process 5 -> Process 6 : ELECTION(5)
Process 6 -> Process 2 : ELECTION(6)
Process 2 -> Process 3 : ELECTION(6)
Process 3 -> Process 5 : ELECTION(6)
Process 5 -> Process 6 : ELECTION(6)

Process 6 receives its own ID back.
***** PROCESS 6 BECOMES COORDINATOR *****

===== COORDINATOR ANNOUNCEMENT =====

Process 6 -> Process 2 : ELECTED(6)
Process 2 -> Process 3 : ELECTED(6)
Process 3 -> Process 5 : ELECTED(6)
Process 5 -> Process 6 : ELECTED(6)

===== ELECTION COMPLETED =====
```

Thank You